CS 2490 – Homework 2

Turn in by 2026/9/5

You **must** turn in your assignment as a single zip file entitled: *lastname_firstname_hw2.zip*. It will contain a pdf file *lastname_firstname_hw2.pdf* with your written answers (for which you can use LaTeX, etc. or you can write your solutions by hand and scan them), and a JSON file for each relevant exercise named *lastname_firstname_hw2_exX.json* (where *X* is the exercise number).

Exercise 1. Draw a DFA that accepts the language of words over the alphabet $\{a, b, c\}$ that contain at least two as and a b (not necessarily in that order).

Use the DFA visualizer online to test your solution, and then export your DFA as a JSON file and include it in your submission.

Exercise 2. Draw a DFA that accepts the language of words over the alphabet $\{a, b\}$ that contain exactly one a.

Prove *properly*[1] that your DFA accepts indeed this language. (*Hint: give meaningful names to your states, and prove that the computation on a given word ends in a state exactly when the word has the property described by the name of that state.*)

[1] As we did in class.

Use the DFA visualizer online to test your solution, and then export your DFA as a JSON file and include it in your submission.

Exercise 3. Draw a DFA that accepts the following language of words over the alphabet $\{a, b\}$:

$$L = \{aba^{2n}ba \mid n \geq 0\}.$$

Use the DFA visualizer online to test your solution, and then export your DFA as a JSON file and include it in your submission.

Exercise 4. Draw a DFA that accepts the following language of words over the alphabet $\{0, 1\}$:

$$L = \{vwv \mid v, w \in \{0, 1\}^* \wedge |v| = 2\}.$$

Use the DFA visualizer online to test your solution, and then export your DFA as a JSON file and include it in your submission.

Exercise 5. Let Σ be any alphabet. We define the *truncation* of a non-empty word w, and we write w^-, to be the same word without its last letter. So if w is written $w_1 w_2 \dots w_n$ (with each $w_i \in \Sigma$), we have

$$w^- = w_1 w_2 \dots w_{n-1}.$$

Let $\mathcal{A}$ be any DFA over Σ such that $\mathrm{L}(\mathcal{A}) \not\ni \varepsilon$ (i.e., the empty word is not recognized by $\mathcal{A}$). Define a DFA[2] that accepts exactly all the truncations of the words in $\mathrm{L}(\mathcal{A}) \not\ni \varepsilon$. In other words, your DFA should accept the language $\{w^- \in \Sigma^* \mid w \in \mathrm{L}(\mathcal{A})\}$.

Once you have defined your DFA, try to prove *properly*[3] that it accepts the language you claim.

To build intuition, I recommend that you take one of the DFA seen in class/exercise session, enter it in the DFA visualizer online, and then play with it to accepts the truncations of the words it originally accepts. This will help you understand what your generic construction should be.

[2] That is, give its alphabet, its set of states, its initial state, its set of final states, and its transition function.

[3] Like we did in class a couple of times.